FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2019

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01– 07, 19 – 25, 37 - 43) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 mark*** *for wrong answer*

  Partial Marks **+1** for each correct option provided no incorrect options is selected.

  Section-A (08 – 13, 26 – 31, 44 - 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S).*

  Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-C (14 – 18, 32 – 36, 50 - 54) *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

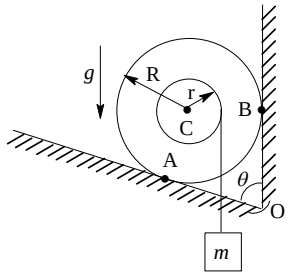
Physics

PART – I

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

1. A massless spool of inner radius r, outer radius R is placed against a vertical wall and a tilted split floor as shown. A light inextensible thread is tightly wound around the spool through which a mass m is hanging. There exists no friction at point A, while the coefficient of friction between the spool and point B is μ. The angle between the two surfaces is ρ.



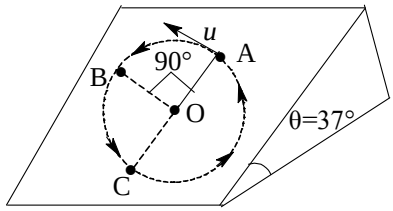
(A) The magnitude of the force on the spool at B in order to maintain equilibrium is $mg\sqrt{(r/R)^2+(1-r/R)^2\ 1/\tan^2\theta}$

(B) The magnitude of the force on the spool at B in order to maintain equilibrium is $mg(1-r/R)1/\tan\theta$

(C) The minimum value of μ for the system to remain in equilibrium is $\cot\theta/(R/r)-1$

(D) The minimum value of μ for the system to remain in equilibrium is $\tan\theta/(R/r)-1$

2. A small sphere of mass m is connected by a string to a nail at O and moves in a circle of radius r on the smooth plane inclined at an angle θ with the horizontal. If the sphere has a velocity u at the top position A. Mark the correct options.



(A) Minimum velocity at A so that the string does not get slack instantaneously is $\sqrt{3/5gr}$

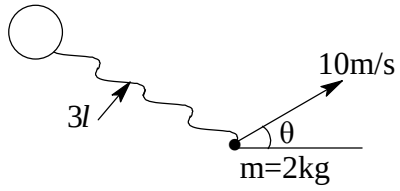
(B) Tension at B is 9mg/5 if the sphere has the required velocity as in option (A).

(C) Tension at C is 18mg/5 in situation of option (B)

(D) None of these

Space for Rough work

3. A string of length $3l$ is connected to a fixed cylinder whose top view is shown in figure. The string is initially slack. The other end of the string (connected to a marble) is moving at a constant velocity of 10 m/s as shown. The string will get stretched at some instant and impulsive tension occurs in the string. If hinge is exerting a force of 40000 N for 0.25 ms on the cylinder to bear up the impact of impulsive tension, then mark the correct statements. (Take string to be light, breaking tension of the string is $2\times10^5\text{N}$)



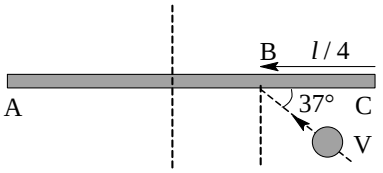
(A) The angle made by the velocity of marble with the length of string when it is just stretched is 60°

(B) The marble will move in a circular path of varying radius with constant speed of $5\sqrt{3}$ m/s, after the string is taut

(C) To answer above two options, the value of θ must be given

(D) The string will break is impulse duration is less than 0.05 min

4. A rod AC of length l and mass m is kept on a horizontal smooth plane. It is free to rotate and move. A particle of same mass m moving on the plane with velocity v strikes the rod at point B making angle 37° with the rod. The collision is elastic. After collision.



(A) The angular velocity of the rod will be 72/55 v/l

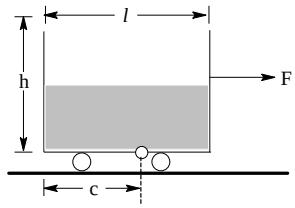
(B) The centre of the rod will travel a distance $\pi l/3$ in the time in which it makes half rotation

(C) Impulse of the impact force is $24\text{m v}/55$

(D) None of these

Space for Rough work

5. A rectangular vessel of dimension $(l \times b \times h)$ and mass M contains a liquid of density ρ. The vessel has an orifice at its bottom at a distance c from the rear wall as shown in the figure.



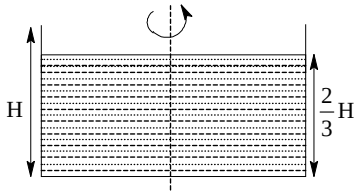
(A) The maximum volume of the water that can be stored when the vessel is accelerated is $hcb/2$

(B) The maximum volume of the water that can be stored when the vessel is accelerated is $hlb/2$

(C) Force F that must be applied when maximum water stored is $\left[M + hcb\rho/2\right]hg/c$

(D) Force F that must be applied when maximum water stored is $\left[M + hcb\rho/2\right]lg/c$

6. A circular cylinder of radius R and height H is filled with water to a height $\left(\frac{2}{3}\right)H$. It starts rotating about its axis with constantly increasing angular speed. Choose the correct alternatives.



(A) At all speeds, shapes of the free surface is parabolloid

(B) The free surface touches first the brim of the cylinder and then the base of the cylinder.

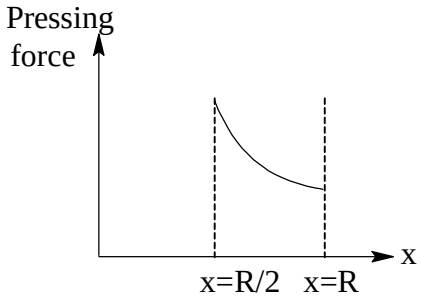
(C) The free surface cannot touch the base without spilling water.

(D) The free surface touches the brim as well as the base at the same instant.

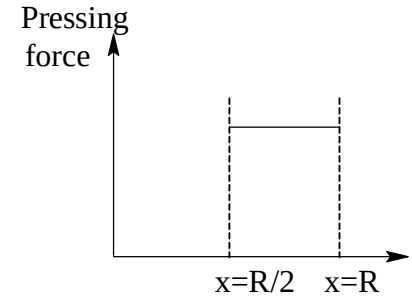
Space for Rough work

7. A tunnel is dug along a chord of the earth at a perpendicular distance R/2 from the earth's centre. The wall of the tunnel may be assumed to be frictionless. A particle is released from one end of the tunnel. The pressing force by the particle on the wall, and the acceleration of the particle vary with x (distance of the particle from the centre) according to:

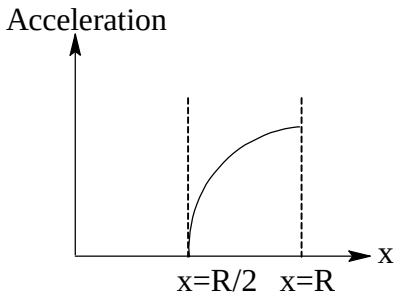
(A)



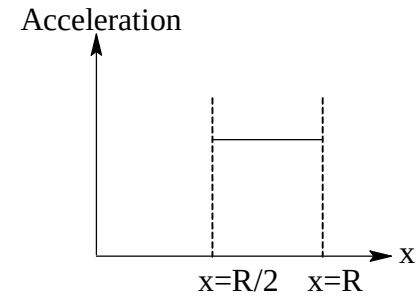
(B)



(C)



(D)



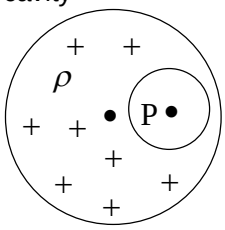
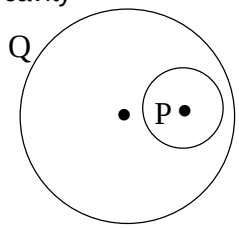
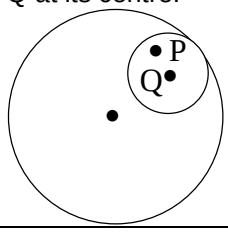
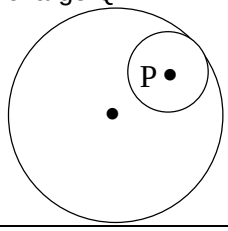
Space for Rough work

(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

Column – 1 gives the continuous charge arrangement, Column – 2 gives the electric field at any arbitary point P inside the cavity and Column – 3 gives the potential at the same arbitary point P:

	Column –1		Column –2		Column –3
(I)	Uniformly charged non-conducting sphere having a cavity  charge density ρ	(i)	$\vec{E}$ in non-uniform	(P)	Potential is constant at every point
(II)	A charged conductor having cavity  Charge conducting sphere	(ii)	E is uniform and non zero	(Q)	Potential is zero at every point
(III)	A uncharged conductor with a V cavity having spherical Q at its centre. 	(iii)	$\vec{E}$ is unidirectional at every point with varying magnitude	(R)	Potential is varying at every point
(IV)	A conducting sphere having charge Q 	(iv)	$\vec{E}$ is zero	(S)	Potential cannot be defined

8. Which of the following is correct combination if the given matrix?

(A) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{R})$ (B) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{P})$

(C) $(\mathrm{I})\ (\mathrm{iv})\ (\mathrm{P})$ (D) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{P})$

9. Which of the following is correct combination if the given matrix?

(A) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{P})$ (B) $(\mathrm{II})\ (\mathrm{iv})\ (\mathrm{P})$

(C) $(\mathrm{I})\ (\mathrm{iv})\ (\mathrm{P})$ (D) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{P})$

10. Which of the following is correct combination if the given matrix?

(A) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{P})$ (B) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{P})$

(C) $(\mathrm{III})\ (\mathrm{i})\ (\mathrm{R})$ (D) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{P})$

Answer 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

For one mole of a monoatomic gas match the following:

	Column –1 [Thermodynamics Process]		**Column –2 [Bulk modulus]**		**Column –3 [Slope of P-V graph]**
(I)	Isothermal	(i)	Infinity	(P)	$-\frac{RT}{V^2}$
(II)	Adiabatic	(ii)	$\frac{5RT}{3V}$	(Q)	Zero
(III)	Isobaric	(iii)	$\frac{RT}{V}$	(R)	Infinity
(IV)	Isochoric	(iv)	Zero	(S)	$-\frac{5RT}{3V^2}$

11. Pick correct combination for the given matrix:

(A) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{P})$ (B) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{P})$

(C) $(\mathrm{I})\ (\mathrm{iv})\ (\mathrm{P})$ (D) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{P})$

Space for Rough work

12. Pick correct combination for the given matrix:

(A) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{S})$ (B) $(\mathrm{II})\ (\mathrm{i})\ (\mathrm{S})$

(C) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{P})$ (D) $(\mathrm{III})\ (\mathrm{ii})\ (\mathrm{S})$

13. Pick correct combination for the given matrix:

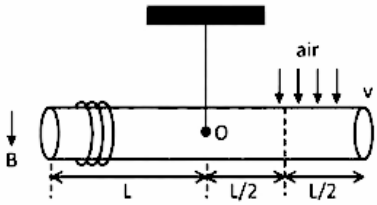
(A) $(\mathrm{IV})\ (\mathrm{ii})\ (\mathrm{R})$ (B) $(\mathrm{IV})\ (\mathrm{iii})\ (\mathrm{R})$

(C) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{R})$ (D) $(\mathrm{IV})\ (\mathrm{iv})\ (\mathrm{R})$

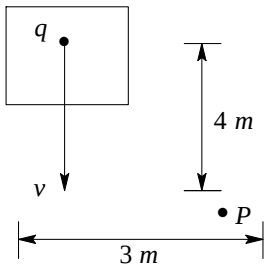
SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. A non-conducting non-magnetic rod having circular cross section of radius R is suspended from a rigid support as shown in figure. A light and small coil of 300 turns is wrapped tightly at the left end of the rod where uniform magnetic field B exists in vertically downward direction. Air of density ρ hits the half of the right part of the rod with velocity V as shown in the figure. What should be current in clockwise direction (as seen from O) in the coil so that rod remains horizontal? Give answer in mA. Given $\frac{2}{\mathrm{LV}}\sqrt{\frac{\pi PB}{\rho}}=\frac{1}{\sqrt{5}}\mathrm{A}^{-1/2}$ [Assume air stops after collision]



15. An elevator carrying a charge of $0.5C$ is moving down with a velocity of $5\times10^{3}ms^{-1}$. The elevator is $4m$ from the bottom and $3m$ horizontally from P as shown in fit.2.202. What magnetic field (in μT) does it produce at point P?

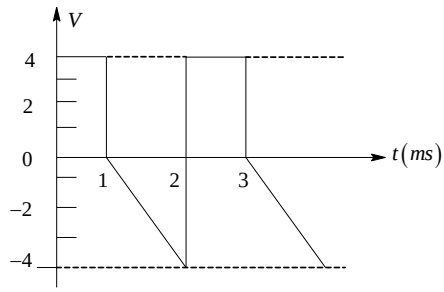


Space for Rough work

16. A long coaxial cable consists of two thin-walled conducing cylinders with inner radius $2\,cm$ and outer radius $8\,cm$. The inner cylinder carries a steady current $0.1A$, and the outer cylinder provides the return path for that current. The current produces a magnetic field between the two cylinders. Find the energy stored in the magnetic field for length $5\,m$ of the cable. Express answer in nJ (use $\ln 2 = 0.7$).

17. A capacitor of capacity $2\mu F$ is charged to a potential difference of $12V$. It is then connected across an inductor of inductance $6\mu H$. What is the current (in A) in the circuit at a time when the potential difference across the capacitor is $6.0V$?

18. Variation of voltage with time is shown in Figure. The rms voltage is found to be $N\sqrt{\frac{2}{3}}V$. Find N.



Space for Rough work

Chemistry

PART – II

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

19. Which of the following process is/are associated with change of hybridization of underlined compounds?
(A) $\underline{Al}(OH)_3$ ppt. dissolved in NaOH
(B) $\underline{B}_2H_6$ is dissolved in THF
(C) $\underline{Si}F_4$ vapours is passed though liquid HF
(D) Solidification of $\underline{P}Cl_5$ vapour.

20. Which of the following statement is/are true? (lp – lone pair and Bp-bond pair)
(A) At $102°$ bond angle in H_2O, $\ell p\text{-}\ell p<\ell p\text{-}Bp<Bp\text{-}Bp$ (Repulsion)
(B) At $104.5°$ bond angle in H_2O, $\ell p\text{-}\ell p=\ell p\text{-}Bp=Bp\text{-}Bp$ (Repulsion)
(C) At $106°$ bond angle in water, $\ell p\text{-}\ell p>\ell p\text{-}Bp>Bp\text{-}Bp$ (Repulsion)
(D) Maximum number of hydrogen bonds H_2O molecule can form are 3.

21. 40 mL 0.05 M solution of sodium sesquicarbonate dihydrate $(Na_2CO_3.NaHCO_3.2H_2O)$ is titrated against $0.05M$ HCl solution, x mL of acid is required to reach the phenolphthalein end point while y mL of same acid were required when methyl orange indicator was used in a separate titration. Which of the following is/are correct statement?
(A) $y\text{-}x=80$ mL
(B) $y+x=160$ mL
(C) If titration is started with phenol phthalein indicator and methyl orange is added at the end point, $2x$ mL of HCl would be required further to reach the end point.
(D) If same volume of same solution is titrated against 0.10 M NaOH, $x/2$ mL of base would be required.

Space for Rough work

22. The incorrect statements are?
(A) The molar volume of every gas at STP is 22.4L
(B) At critical point, compression factor is unity
(C) A gas with large 'b' has high critical pressure.
(D) For real gas, $\lim_{p\to 0}\left(\frac{\text{PV}}{\text{RT}}\right)=1$ where symbols have their usual meaning.

23. When $0.5\,\text{mL}$ of a $1.0\,\text{M}\,\text{HCl}$ solution is mixed with $5.0\,\text{mL}$ of a $0.1\text{M}\,\text{NaOH}$ solution, temperature of solution increases by $2°\text{C}$. Which of the following(s) can be predicated accurately from this observation?
(A) If 10mL of same HCl is mixed with $10\,\text{mL}$ of same NaOH, temperature rise will be $4°\text{C}$
(B) If 10mL of 0.05 M HCl is mixed with $10\,\text{mL}$ of $0.5\,\text{M}$ NaCl, the temperature rise will be $2°\text{C}$
(C) If 5mL of 0.1 M HCl is mixed with 5 mL 0.1 M NH_3 solution, the temperature rise will be less than $2°\text{C}$
(D) If 5 mL of 0.1 M CH_3COOH is mixed with 5 mL 0.1 M NaOH, the temperature rise will be less than $2°\text{C}$

24. At a certain temperature (T_1), some hydrazine (N_2H_4) gas is placed in a sealed flask where the following equilibrium is established.

$$3\text{N}_2H_4(g)\underset{\leftarrow}{\overset{K_1}{\rightleftharpoons}}4NH_3(g)+N_2(g) \qquad \Delta H°>0$$

If the temperature is increased to T_2, some ammonia gas also decomposes as

$$2NH_3(g)\underset{\leftarrow}{\overset{k_2}{\rightleftharpoons}}N_2(g)+3H_2(g) \qquad \Delta H°>0$$

What would be observed as temperature is increased from T_1 to T_2?
(A) Some more hydrazine would decompose.
(B) If $\text{K}_1<\text{K}_2$, concentration of NH_3 would decrease.
(C) Partial pressure of N_2 would be doubled.
(D) All of these would be observed.

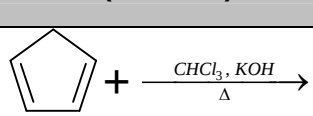
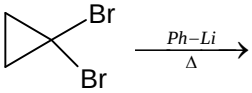
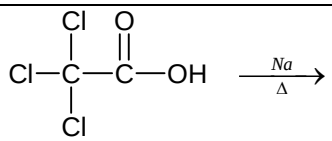
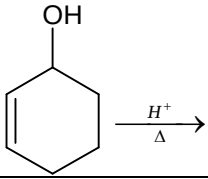
25. The yellow precipitate is obtained when:
(A) Lead acetate solution is treated with K_2CrO_4
(B) $\text{Pb}(\text{NO}_3)_2$ solution is treated with K_2CrO_4
(C) AgNO_3 solution is treated with KI
(D) H_2S is passed through a solution of CdSO_4

Space for Rough work

(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

	Column –1 (Reaction)		Column –2 (Intermediate)		Column –3 (Double bond equivalent in product)
(I)		(i)	Carbocation	(P)	0
(II)		(ii)	Carbonion	(Q)	1
(III)		(iii)	Free Radical	(R)	2
(IV)		(iv)	Carbene	(S)	3

26. Which of the following is correct combination if the given matrix?
(A) $(\mathrm{II})\ (\mathrm{iv})\ (\mathrm{R})$ (B) $(\mathrm{I})\ (\mathrm{ii})(\mathrm{iv})\ (\mathrm{R})$
(C) $(\mathrm{II})\ (\mathrm{ii})(\mathrm{iv})\ (\mathrm{R})$ (D) $(\mathrm{III})\ (\mathrm{iii})\ (\mathrm{P})$

27. Which of the following is correct combination if the given matrix?
(A) $(\mathrm{IV})(\mathrm{i})\ (\mathrm{R})$ (B) $(\mathrm{I})\ (\mathrm{iv})\ (\mathrm{R})$
(C) $(\mathrm{III})\ (\mathrm{iv})\ (\mathrm{P})$ (D) $(\mathrm{I})\ (\mathrm{ii})(\mathrm{iv})\ (\mathrm{S})$

28. Which of the following is used for extraction of copper?
(A) $(\mathrm{IV})(\mathrm{i})\ (\mathrm{R})$ (B) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{Q})$
(C) $(\mathrm{III})\ (\mathrm{ii})(\mathrm{iv})\ (\mathrm{P})$ (D) $(\mathrm{III})\ (\mathrm{iv})\ (\mathrm{Q})$

Space for Rough work

Answer 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

	Column –1		Column –2		Column –3
(I)	$2NH_3(g) \rightleftharpoons N_2(g)+3H_2(g)$	(i)	Introduction of $H_2(g)$ at constant volume shift equilibrium back.	(P)	$Kp=K_C(RT)$
(II)	$PCl_5(g) \rightleftharpoons PCl_3(g)+Cl_2(g)$	(ii)	α increases on increasing temperature	(Q)	$Kp=K_C(RT)^2$
(III)	$2HI(g) \rightleftharpoons H_2(g)+I_2(g)$	(iii)	On increasing the pressure, concentration of all the reactants increases	(R)	$Kp=K_C(RT)^{-2}$
(IV)	$NH_4HS(s) \rightleftharpoons NH_3(g)+H_2S(g)$	(iv)	α is independent of equilibrium pressure	(S)	$Kp=K_C$

29. Which of the following is correct combination?
(A) $(III)\ (i)(iii)(iv)\ (S)$ (B) $(III)\ (i)(ii)(iv)\ (S)$
(C) $(IV)\ (ii)(iii)\ (Q)$ (D) $(IV)\ (ii)(iv)\ (Q)$

30. Which of the following is correct combination?
(A) $(I)\ (i)(ii)(iii)(iv)\ (Q)$ (B) $(III)\ (i)(ii)(iii)(iv)\ (S)$
(C) $(III)\ (i)(ii)(iii)\ (S)$ (D) $(I)\ (ii)(iii)(iv)\ (Q)$

31. Which of the following is correct combination?
(A) $(I)\ (i)(ii)(iii)\ (Q)$ (B) $(II)\ (ii)(iii)(iv)\ (P)$
(C) $(II)\ (i)(ii)(iii)\ (P)$ (D) $(I)\ (iii)(iv)\ (Q)$

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. The solubility product constant of a metal carbonate MCO_3 is 2×10^{-12} at 25°C. A solution is 0.1 M in $M(NO_3)_2$ and it is saturated with 0.01M CO_2. Also the ionization constant of CO_2 are $Ka_1 = 4\times10^{-7}$ and $Ka_2 = 5\times10^{-11}$ at 25°C. The minimum pH that must be maintained to start any precipitation is?

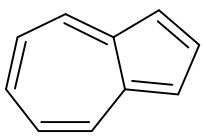
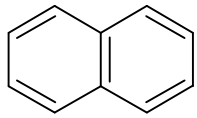
33. 2.0 moles of an ideal gas of unknown heat capacity are in a piston fitted cylinder at 300K. The piston has a mass on the top of it that supplies a constant pressure. The piston is then brought into contact with a thermostal. Heat flow into the system causing the gas to both expand and change temperature. After the temperature has equilibrated, the work done by the gas on surrounding was found to be 3.32 kJ. Also 14.94 kJ of heat flows into the system. What is the atomicity of the gas. Assume $R = 8.3\ jK^{-1}\ mol^{-1}$.

34. The standard reduction potential of silver chloride electrode (metal – sparingly soluble salt electrode) is 0.209 V and for silver electrode is 0.80 V. If the moles of AgCl that can dissolve in 10 L of a 0.01 M NaCl solution is represented as 10^{-y}, then find the value of y?

35. Consider the following reversible system:

$$A(g) + 2B(g) \rightleftharpoons AB_2(g) \qquad K_C = \frac{1}{4}$$

The above equilibrium is established in a 1.0 L flask and at equilibrium 2 moles of each A and B are present. If 2 moles of B is added further, how many moles of AB_2 should be added so that moles of A does not change?

36. Which of these statements are correct?
(i) 1, 3, 5-cycloheptatriene is not acidic while 1, 3-cyclopentadiene is acidic in nature.
(ii) p-chlorophenol is less acidic than p-fluorophenol
(iii) $-N=O$ shows both +M and -M effect.
(iv) singlet halo-carbene are more stable than triplet halo-carbene
(v) conjugated acid of pyridine remains aromatic but pyrrole does not remain aromatic.
(vi) Dipole moment of  is greate than 
(vii) energy needed for homolytic bond fission is less than that required for the heterolytic bond fission.

Space for Rough work

Mathematics

PART – III

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

37. Let $x, y \in [0, 2\pi]$ and satisfying the equation $\sin^3 x + \cos^3 y + 6\sin x \cos y = 8$. If $\ell = x + y$ and $m = x - y$, then:

(A) least value of ℓ is $\frac{\pi}{2}$ (B) largest value of ℓ is $\frac{5\pi}{2}$

(C) least value of m is $\frac{\pi}{2}$ (D) largest value of m is $\frac{\pi}{2}$

38. If $\sin^{-1}\left(\tan\left(\frac{\pi}{2}\sin\theta\right)\right) + \cos^{-1}\left(\cot\left(\frac{\pi}{2}\cos\theta\right)\right) = \frac{\pi}{2}$, then value of $\sin\frac{\theta}{3}$ can be:

(A) 0 (B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

39. Let α, β *and* γ be three real numbers. Suppose that $\cos\alpha + \cos\beta + \cos\gamma = 1$ and $\sin\alpha + \sin\beta + \sin\gamma = 1$, then smallest possible value of $\cos\alpha = \frac{-p - \sqrt{q}}{r}$

(where p, q, r $\in \mathbb{Z}^+$)

(A) $p + q + r = 12$ (B) $p + q + r = 10$

(C) $|p + q - r| = 4$ (D) $|p - q + r| = 2$

Space for Rough work

40. Consider an equation $8x^4 - 16x^3 + 16x^2 - 8x + a = 0$ then the sum of all the non-real roots of the equation can be $[a \in \mathbb{R}]$

(A) 1 (B) 2

(C) $\frac{1}{2}$ (D) None of these

41. The vector(s) which is/are coplanar with vectors $\hat{i} + \hat{j} + 2\hat{k}, \hat{i} + 2\hat{j} + \hat{k}$ and perpendicular to the $\vec{c}$, where $\vec{c}$ is unit vector perpendicular to the plane of vectors of vectors $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 3\hat{k}$ is/are:

(A) $3\hat{i} + 4\hat{j} + 5\hat{k}$ (B) $\frac{3}{\sqrt{2}}\hat{i} + 2\sqrt{2}\hat{j} + \frac{5}{\sqrt{2}}\hat{k}$

(C) $-\left(\hat{i} + \frac{4}{3}\hat{j} + \frac{5}{3}\hat{k}\right)$ (D) $\sqrt{3}\hat{i} + \frac{4}{\sqrt{3}}\hat{j} + 5\sqrt{3}\hat{k}$

42. On xy-plane, the path defined by the equation $\frac{1}{x^m} + \frac{1}{y^m} + \frac{k}{(x+y)^n} = 0$ is

(A) a parabola if $m = \frac{-1}{2}, k = -1, n = 0$ (B) a hyperbola if $m = 1, k = -1, n = 0$

(C) a pair of lines if $m = k = n = 1$ (D) a pair of lines if $m = k = -1, n = 1$

43. Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ be a matrix if $A^{10} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then

(A) $a + d$ is a multiple of 13 (B) b is an even integer

(C) Number of factors of $d - b$ is 11 (D) Number of factors of $a + b + c + d$ is 22

Space for Rough work

(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

Column – 1: Real valued function
Column – 2: Domain of the function
Column – 3: Range of the function
Match the following column(s) (where [.] denotes greatest integer function and {.} denotes fractional part function)

	Column –1		Column –2		Column –3
(I)	$f_1(x)=\left[\log_{\sqrt{2}}\sqrt{\left\|(x-1)^3\right\|+\left\|(x-3)^3\right\|}\right]$	(i)	Number of integers in the domain of f(x) is 2	(P)	Set of all non positive integers $\cup\{1\}$
(II)	$f_2(x)=\left[\log_{\sqrt{2}}\left(\sin\ln\left(\frac{\sqrt{4-x^2}}{1-x}\right)+\cos\ln\left(\frac{\sqrt{4-x^2}}{1-x}\right)\right)\right]$	(ii)	Number of integers not in the domain of f(x) is 2	(Q)	Set of all non-negative integers
(III)	$f_3(x)=\left\{\frac{\left\|x+5\right\|-2}{x^2+10x+21}\right\}$	(iii)	Number of integers in the domain of f(x) is 3	(R)	Set of all positive integers
(IV)	$f_4(x)=\left[\frac{1}{\sqrt{\ln\left(\cos^{-1}x\right)}}\right]$	(iv)	Number of integers in the domain of f(x) is infinite	(S)	$\left(0,\frac{1}{2}\right]$

44. Which of the following combination is correct?
(A) $(\mathrm{I})\ (\mathrm{iv})\ (\mathrm{P})$ (B) $(\mathrm{I})\ (\mathrm{iv})\ (\mathrm{R})$
(C) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{R})$ (D) $(\mathrm{III})\ (\mathrm{i})\ (\mathrm{S})$

45. Which of the following combination is correct?
(A) $(\mathrm{IV})(\mathrm{i})\ (\mathrm{P})$ (B) $(\mathrm{III})\ (\mathrm{ii})\ (\mathrm{S})$
(C) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{Q})$ (D) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{S})$

46. Which of the following combination is correct?
(A) $(\mathrm{II})(\mathrm{iv})\ (\mathrm{P})$ (B) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{P})$
(C) $(\mathrm{II})\ (\mathrm{i})\ (\mathrm{P})$ (D) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{R})$

Space for Rough work

Answer 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

If three planes are given

$P_1 : 2x + 3y + 4z = 9$

$P_2 : 4x + 5y + 6z = 15$

$P_3 : y + az = b$

And two lines are given

$L_1 : \vec{r} = \left(\hat{i} + a_1\hat{j} + \hat{k}\right) + \lambda\left(b_1\hat{i} + b_2\hat{j} + \hat{k}\right)$

$L_2 : \vec{r} = \mu\left(\hat{i} + \hat{j} + \hat{k}\right)$

Where line L_1 is contained in P_1, P_2 and P_3:

Then,

	Column –1		Column –2		Column –3
(I)	$b_1 - b_2$ is -3	(i)	Acute angle between P_1 & P_3 is $\cos^{-1}\left(\frac{11}{\sqrt{145}}\right)$	(P)	Image of point $(0,0,0)$ w.r.t. P_3 is $\left(0, \frac{6}{\sqrt{5}}, \frac{12}{\sqrt{5}}\right)$
(II)	$a_1 + a$ is 3	(ii)	Acute angle between L_2 & P_3 is $\cos^{-1}\left(\sqrt{\frac{3}{5}}\right)$	(Q)	Image of point $(0,0,0)$ w.r.t. line L_1 is $(1,1,1)$
(III)	$b + b_1 + b_2$ is 2	(iii)	Distance between L_1 and L_2 is zero	(R)	Distance of point $(0,0,0)$ from P_1 measured parallel to L_2 is $\frac{9}{\sqrt{29}}$
(IV)	$\int_{b_2}^{b_1} \{x\}\, dx$ is $\frac{1}{2}$ (where {.} represents fractional part function)	(iv)	Acute angle between L_2 & P_1 is $\sin^{-1}\left(\frac{3\sqrt{3}}{\sqrt{29}}\right)$	(S)	Distance of a point $(1,0,0)$ from P_3 measured parallel to L_2 is $\sqrt{3}$

47. Which of the following is correct?

(A) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{Q})$ (B) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{P})$

(C) $(\mathrm{III})\ (\mathrm{ii})\ (\mathrm{S})$ (D) $(\mathrm{IV})\ (\mathrm{iv})\ (\mathrm{R})$

Space for Rough work

48. Which of the following is correct?

(A) $(\text{III})\ (\text{iv})\ (\text{S})$ (B) $(\text{III})\ (\text{ii})\ (\text{P})$

(C) $(\text{IV})\ (\text{i})\ (\text{P})$ (D) $(\text{II})\ (\text{iii})\ (\text{R})$

49. Which of the following is incorrect?

(A) $(\text{II})\ (\text{i})\ (\text{P})$ (B) $(\text{III})\ (\text{iii})\ (\text{S})$

(C) $(\text{III})\ (\text{iv})\ (\text{R})$ (D) $(\text{III})\ (\text{iii})\ (\text{P})$

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. A special deck of cards contains 49 cards, each labeled with a number from 1 to 7 and coloured with one of seven colours. Each number – colour combination appears on exactly one card. Mokshika will select a set of eight cards from the deck at random. Given that she gets atleast one card of each colour and atleast one card with each number, the probability that Mokshika can discard one of her cards and still have atleast one card of each colour and atleast one card with each number is $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $|p-q|$:

51. If $f(y)=\lim\limits_{n\to\infty}\sum\limits_{n=1}^{n}\frac{e^{n+y}(e-1)}{(e^n-1)(e^{n+1}-1)}$ then the value of $\int\limits_0^1 f(x)dx$ is:

52. Let $y=f(x)\,(f:\mathbb{R}\to\mathbb{R})$ be an explicit function defined by the implicit equation $x^3+y^3+3(x^2+y^2)+3(x+y)=14$ and g be the inverse of f. If $\frac{d}{dx}\left(f(x+g(x)).g(x+f(x))\right)$ at $x=-1+\sqrt[3]{15}$ is equal to $\lambda(15)^{2/3}$, where $\lambda\in\mathbb{Z}$, then find the value of $|\lambda|$.

53. The remainder when $2^{\frac{1.2}{2}}+2^{\frac{2.3}{2}}+2^{\frac{3.4}{2}}+.....+2^{\frac{2011.2012}{2}}$ Is divided by 7 is_____

54. Evaluate: $\left[x^2\left(\frac{\cos\left(\cot^{-1}x\right)+\frac{1}{x}\left(\sin\left(\cot^{-1}x\right)\right)}{\cot\left(\cos ec^{-1}x\right)+\tan\left(\cos ec^{-1}x\right)}\right)+\frac{1}{x^4}\right]^{\frac{1}{2}}=$

Space for Rough work